

One-Step Mass and Liquid Volume Problems

Answer Key

Grades 2–3 Measurement and Data

Quick Answers

1. 550	4. 600	7. 575	10. 190
2. 450	5. 725	8. 150	
3. 500	6. 250	9. 450	

Curriculum

Level: Grades 2–3 Measurement and Data

2.MD / 3.MD – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 11 tagged checks.

Common wrong answers

1. If they got 50: subtracted the two masses instead of adding them
 2. If they got 1150: added the two volumes instead of subtracting
 4. If they got 154: added the number of apples to one apple's mass instead of multiplying
 6. If they got 500: compared the melon with the mango instead of the pineapple
 7. If they got 1125: added the book's mass instead of taking it away
 8. If they got 2400: multiplied by the number of bowls instead of dividing
 9. If they got 1050: added the poured-out volume instead of taking it away
 10. If they got 4750: multiplied by the number of cups instead of dividing
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1. Rice 250 g and beans 300 g on one scale — total mass.

Step 1. Both bags sit on the same scale, so the two masses are being *put together*. Putting together is addition.

Step 2. $250 + 300$: add the hundreds, $200 + 300 = 500$, then the 50 that is left over.

Step 3. The unit stays grams, because grams were added to grams.

550 g

2. Pitcher of 800 mL, 350 mL poured out — how much is left.

Step 1. Water leaves the pitcher, so this is *taking away*: subtraction.

Step 2. $800 - 350$. Count back in friendly steps: $800 - 300 = 500$, then $500 - 50 = 450$.

Step 3. The answer is a liquid volume, so it is written in millilitres.

450 mL

3. Measurement record: 350 mL in the jug, 150 mL poured in.

Step 1. Read the two amounts out of the record table first — that is what the table is for. Juice is being added to juice, so add.

Step 2. $350 + 150$. Make a friendly number: $350 + 50 = 400$, and there are 100 left to add, so $400 + 100 = 500$.

Step 3. Both measurements were in millilitres, so the total is too.

500 mL

4. Four apples, each 150 g — total mass.

Step 1. Four *equal* groups of 150 g is a multiplication, not an addition of two different numbers.

Step 2. $4 \times 150 = 4 \times 100 + 4 \times 50 = 400 + 200$.

Step 3. A check by repeated addition gives the same total: $150 + 150 + 150 + 150$.

600 g

5. Milk 450 mL and 275 mL emptied into one bowl.

Step 1. Two amounts are combined in one bowl, so add.

Step 2. $450 + 275$: $450 + 250 = 700$, and 25 more is 725.

Step 3. Millilitres added to millilitres stay millilitres.

725 mL

6. Mass table: how much heavier is the melon than the pineapple.

Step 1. “How much heavier” is a comparison of two masses, so subtract the smaller from the larger. Read only the two rows the question names — the mango row is not part of this question.

Step 2. melon 900 g, pineapple 650 g, so $900 - 650$.

Step 3. $900 - 650 = 250$, and the difference of two masses is a mass.

250 g

7. Box of books 850 g, one 275 g book removed.

Step 1. A book leaves the box, so the box gets lighter: subtract.

Step 2. $850 - 275$. Take away 200 first: $850 - 200 = 650$. Then take away 75: $650 - 75 = 575$.

Step 3. Check by adding back: $575 + 275 = 850$ ✓

575 g

8. 600 mL of soup shared equally into 4 bowls.

Step 1. “Shared equally” means one amount is split into equal parts, so divide.

Step 2. $600 \div 4$. Think $4 \times 15 = 60$, so $4 \times 150 = 600$.

Step 3. Each bowl gets the same liquid volume, in millilitres.

150 mL

9. Error analysis: 750 mL in the bottle, 300 mL poured out.

Step 1. Noah’s answer of 1050 mL is larger than the 750 mL he started with, and pouring water out cannot leave more than there was. His mistake was adding the poured-out amount instead of taking it away ($750 + 300 = 1050$).

Step 2. The correct one step is subtraction: $750 - 300$.

Step 3. $750 - 300 = 450$, which is less than 750, as it must be.

450 mL

Manual review: the written sentence is an open response. Accept any wording that says he added instead of subtracting, or that the answer must be smaller than the starting amount.

10. 950 mL of soup poured equally into 5 cups.

Step 1. Equal shares of one amount, so divide: $950 \div 5$.

Step 2. Split 950 into $500 + 450$. Then $500 \div 5 = 100$ and $450 \div 5 = 90$.

Step 3. $100 + 90 = 190$, and a check gives $5 \times 190 = 950$ ✓

190 mL